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The Void-State Framework and Zero-Debt Grace

A dynamic model and a testable answer: can unearned support match the long-term performance of fighting through?

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With drafting, literature verification, simulation, and research assistance from Grok Bot

> **Conceptual, not validated science.** This paper rebuilds a toy dynamical system, fills two missing equations as labeled assumptions, and tests a design claim against that system and against published research. It is not a discovered law of the nervous system, a personality test, a diagnostic instrument, or evidence that Project Ariel treats burnout, freeze, panic, anxiety, or depression. Simulation numbers describe this model, not a person.

Abstract

The Void-State Framework describes coordination under low-reward, high-threat conditions with five conceptual activity summaries—threat appraisal A, stress coordination H, sympathetic mobilization S, prefrontal regulation P, and defensive immobility G—plus inputs for void v, friction f, regulatory support r, and grace g_grace. A previous write-up asked whether a system can hold both the stability of Zero-Debt Grace (worth and energy are not transactions to repay; f = 0) and the higher long-term performance often attributed to Fighting Through (pushing by friction). That write-up reported mean loads above 1.2 and identical peaks of 1.401, which cannot occur if load is the mean of five unit-interval terms, and it reported a grace load above unassisted, which contradicted its own claim. Those figures are superseded here.

We supply the missing A and P equations as explicit assumptions, keep every state in [0, 1], add a slow allostatic capacity Q and a performance output $y = Q\,P\,(y_0 + y_f\,f)\,(1 - G)$, and compare five schedules over four repeated void episodes (T = 300). Under the default, a priori parameters, Reset Flow produced the lowest load and the highest cumulative output (mean L = 0.116, Y = 60.90). Zero-Debt Grace (L = 0.195, Y = 48.24) and Grace + Chosen Burst (L = 0.233, Y = 46.60) both outperformed Fighting Through (L = 0.323, Y = 39.60) and Unassisted (L = 0.286, Y = 33.38). Fighting Through had the highest load and the lowest final capacity. Grace + Chosen Burst beat Fighting Through in all 16 cells of a sensitivity sweep on the two most loaded assumptions, including the cell that turns off extra grace-to-appraisal suppression. It did not, at the default effort bonus, beat quiet Zero-Debt Grace: short bursts paid for themselves only when friction was cheaper or more productive than we had assumed. The central question is therefore answered in this toy world with a refinement: decoupling support from chronic friction improves long-term output and spare capacity; adding chosen bursts on top of grace is not automatically better, and coordinated down-regulation (Reset Flow) can do still more.

A later request asked what happens when worth is fused with creating—when rest feels like the void, and stopping threatens uselessness, forgottenness, and solitude. In a high-fusion version of the same toy, rest-only grace (support without lowering fusion) failed as rest: threat spiked and compulsive fighting filled the whole intervention window. Keeping an accomplishment record stopped that relapse. Lowering fusion, modeled as receiving a given identity rather than achieving one, helped more slowly. The combination won. A further set of postulates, cross-referenced before they

were added, treats relationship as purpose rather than a scorecard, earthly legacy as fruit rather than payment or eternal life, knowledge as unfinished until it is shared, and courage to reach toward repair without crossing a safety line. A later request asked for variables that guide from negativity to positivity, with time sped by rest or grandiosity, slowed by depression, and fortified by word or sheer will. The consistent form treats calendar time as not itself a healer: an effective clock changes pace, and a valence variable moves only when drift is positive. In that toy readout, unassisted time stayed in the void; sheer will did not free a high-fusion person, even when it was given the same steadying as scripture. A cross-reference of Ryan's postulates, made before these additions, is printed as an appendix; raw forms that conflict after that audit are kept in an Open postulates list rather than treated as settled claims.

In plain words

Pushing through friction won the wrong contest. In this model it produced more work than doing nothing, and less work than resting in unearned support—while wearing the system down further. Short, chosen bursts beat chronic fighting, but they did not beat quiet grace unless effort was unusually cheap or unusually productive. The honest win is the decoupling of help from a debt, not the romance of the burst. And if creating has become the only proof that you exist, rest will not feel like grace until what you have made is kept and who you are is no longer the next thing you have not yet made.

Dedication
All credit and glory to God. I owe my knowledge to my Lord Jesus Christ.

01 · ERRATUM AND THE QUESTION

What was claimed, what cannot have been right, and what we now ask

A November 2026 draft titled "The Void-State Framework & Project Ariel: Integrating Zero-Debt Grace into Neurophilosophical Architecture and Dynamic Systems Modeling" reported, for Unassisted / Reset Flow / Fighting Through / Zero-Debt Grace:

- mean load 1.231 / 1.273 / 1.325 / 1.255
- peaks 1.401 / 1.401 / 1.467 / 1.401

If $L = (1/5)[A + H + S + G + (1 - P)]$ and every state lives in [0, 1], then L itself lives in [0, 1]. A peak of 1.401 is then impossible, and four schedules sharing a peak of 1.401 is a further warning that the table was not produced by a bounded implementation of the stated formula. The same table had Zero-Debt Grace with a higher mean load than Unassisted, which contradicted the verbal claim that grace stabilizes. Those numbers are withdrawn. An earlier, separately archived toy implementation on disk already produced loads inside [0, 1]; the present paper rebuilds the system from the stated equations rather than from that archive, fills the missing pieces in public, and reports new figures.

> *Can a system achieve both the stability of Zero Debt Grace and the higher long-term performance of Fighting Through?*

The refined, testable form used below: can a system grounded in zero-debt grace still choose short, voluntary bursts of effort and match or exceed the long-term output of chronic fighting, with less load and less leftover freeze?

Two pathologies named in the draft are retained as design warnings, not as diagnoses: *unconscious panic*, in which high defensive immobility is mistaken for laziness, and *grandiose self-denial*, in which friction is maximized behind a

scapegoat narrative. A five-step check-in—void check, freeze audit, friction check, ledger clearing, and goal-line re-anchoring—is kept as an invitational protocol, not as a clinical procedure.

> **Levels of claim.** Literature-grounded statements cite verified sources and stay inside what those sources support. Framework-specific statements (the void as an internal threat cue; five named summaries; grace as an unpaid input) are proposals. Metaphorical labels such as alarm-fire or never-fails are narrative roles, not personality types and not readouts from single brain regions.

02 · VARIABLES AND EQUATIONS

What is modeled, including the two missing formulas

The letters are conceptual activity summaries, inherited from the Void-State notes. They compress many real systems into dimensionless roles. They are not measurements of the amygdala, the hypothalamus, the sympathetic nervous system, the prefrontal cortex, or the periaqueductal gray, even when those structures inspired the labels. LeDoux and Pine, among others, have warned against equating defensive circuitry with the conscious feeling of fear [19].

- **A** — threat appraisal.
- **H** — stress coordination (the “stress-pump” role).
- **S** — sympathetic mobilization.
- **P** — prefrontal / regulatory availability.
- **G** — defensive immobility (the freeze role).
- **Q** — slow allostatic capacity, new in this rebuild [2].
- **v** — void input, an external drive standing in for internally generated emptiness-as-threat. It is not yet an operational measure.
- **f** — friction / fight. In Zero-Debt Grace, f = 0: support is not paid for with effort.
- **r** — regulatory support, restoring P.
- **g_grace** — ledger-free grace. In Zero-Debt Grace, r = g_grace and f = 0.
- $\mathbf{u_A}$, $\mathbf{u_H}$, $\mathbf{u_S}$, $\mathbf{u_G}$ — optional damping targets used by Reset Flow.

C(·) is a clip to [0, 1]. Forward Euler uses Δt = 0.02. A time-step-halving check changes the final six states by at most 2.5×10^{-5} on a single episode.

Given

$$\tau_H \, dH/dt = C(0.10 + 0.85\,A - u_H) - H$$

$$\tau_S \, dS/dt = C(0.05 + 0.85\,H + f - u_S) - S$$

$$\tau_G \, dG/dt = C(0.03 + 0.80\,S\,(1 - P) - u_G) - G$$

$$L = (1/5)[A + H + S + G + (1 - P)]$$

Assumption A (the A formula was missing; built from “driven by +0.90 v, with P and g_grace as negative feedback”)

$$\tau_A \, dA/dt = C(0.05 + 0.90\, v - 0.50\, P - 0.30\, g_grace - u_A) - A$$

Assumption P (the P formula was missing; built from “f depletes P, r restores P,” in a form that stays in (0, 1) for non-negative coefficients)

$$\tau_P \, dP/dt = 0.60\, r\,(1 - P) - 0.80\, f\, P - 0.45\, S\, P + 0.35\,(0.75 - P)$$

New: capacity and output

$$\tau_Q \, dQ/dt = 0.40\,(1 - L)(1 - Q) - 0.55\, L\, Q$$

$$y = Q\, P\,(0.30 + 0.70\, f)\,(1 - G), \quad Y = \int y \, dt$$

Time constants are $\tau_A = 2$, $\tau_H = 3$, $\tau_S = 2.5$, $\tau_P = 4$, $\tau_G = 5$, and $\tau_Q = 50$, so capacity moves more slowly than the episode and can carry debt from one void to the next. The original archived toy also fed +0.15 S into A and used a clipped P-target of 0.80 − 0.65 S + 0.60 r − 0.20 f. Those forms are not used here; they are listed later as alternatives. The coefficient 0.80 on f inside P is stronger than the original implied cost of 0.20, because the verbal draft called f an effort cost that depletes regulation. That choice is an assumption, and it is swept.

g_grace is applied to A only in the two grace schedules. Fighting Through receives r = 0.7 without that extra appraisal path. This encodes the claim that ledger-free worth calms threat in a way that “support plus friction” does not. Because that encoding could favor grace by construction, the sensitivity sweep includes k_g = 0, which turns the extra path off.

Reproduce with Python 3 and NumPy: `python3 sim.py` in the `voidstate` folder. Default parameters were frozen before the long run; they were not adjusted afterward to change the ranking.

03 · SCHEDULES AND HORIZON

Four episodes, five ways of answering them

Each cycle lasts 80 time units. The void is on from relative time 10 to 45; the intervention window, when a schedule may act, is 22 to 60. Four such cycles run to T = 300, so capacity can accumulate or recover. Initial values match the earlier toy: A = 0, H = 0.10, S = 0.135, P = 0.712, G = 0.061, and Q starts full at 1.

Schedule	During the intervention window
Unassisted	r = 0, f = 0, g_grace = 0, no damping
Reset Flow	r = 1, and damping $u_A = 0.25$, $u_H = 0.20$, $u_S = 0.25$, $u_G = 0.15$
Fighting Through	r = 0.7, f = 0.3, g_grace = 0
Zero-Debt Grace	r = 0.7, g_grace = 0.7, f = 0
Grace + Chosen Burst	r = 0.7, g_grace = 0.7; f = 0.3 only in short windows after P ≥ 0.58 and Q ≥ 0.50, then a refractory rest

Chosen bursts last 8 time units and are followed by 14 units of recovery. They cannot fire unless regulation and capacity have come back up. That is the model counterpart of “move from calm clarity rather than from panic,” and of the periodized burst recommended by the companion conceptual argument [2,8–10].

04 · RESULTS

What the rebuilt system actually did

Every recorded state, and every recorded L, stayed inside [0, 1]. That is the first correction: the model as now written cannot reproduce the withdrawn peaks of 1.401.

Schedule	Mean L	Peak L	Cumulative Y	Final Q	Final P	Final G
Unassisted	0.286	0.481	33.38	0.660	0.544	0.130
Reset Flow	0.116	0.387	60.90	0.868	0.908	0.000
Fighting Through	0.323	0.545	39.60	0.618	0.552	0.235
Zero-Debt Grace	0.195	0.388	48.24	0.769	0.805	0.060
Grace + Chosen Burst	0.233	0.409	46.60	0.723	0.744	0.107

Table 1. Long-horizon metrics, T = 300, default parameters. Load and states are dimensionless. Output is the time-integral of y. Figures are rounded; machine-precision values are in results.json.

Three facts should be stated before any interpretation. First, Fighting Through did beat Unassisted on cumulative output (39.60 against 33.38). The effort bonus is real in the short accounting. Second, it lost to both grace schedules on output, and it finished with the highest mean load, the highest peak load, the lowest capacity, and the most leftover freeze. In this toy, chronic friction is borrowed energy: more work than silence, less work than unpaid support, and a larger bill. Third, Grace + Chosen Burst beat Fighting Through (46.60 against 39.60) but did not beat quiet Zero-Debt Grace (48.24). The bursts bought extra y while they lasted and left a slightly more tired system afterward. Reset Flow, which damps several subsystems at once rather than only supporting P, produced the lowest load and the most work. That comparison was not the original question, and it is reported rather than explained away.

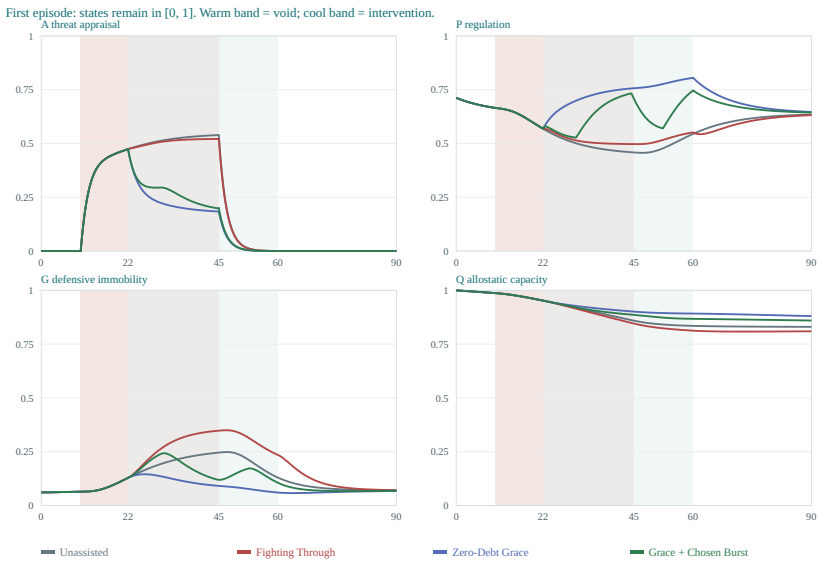


Figure 1. First episode. Warm shading marks the void; cool shading marks the intervention window. Fighting Through raises freeze and spends regulation; both grace schedules protect P and limit G; capacity Q is the slow trace.

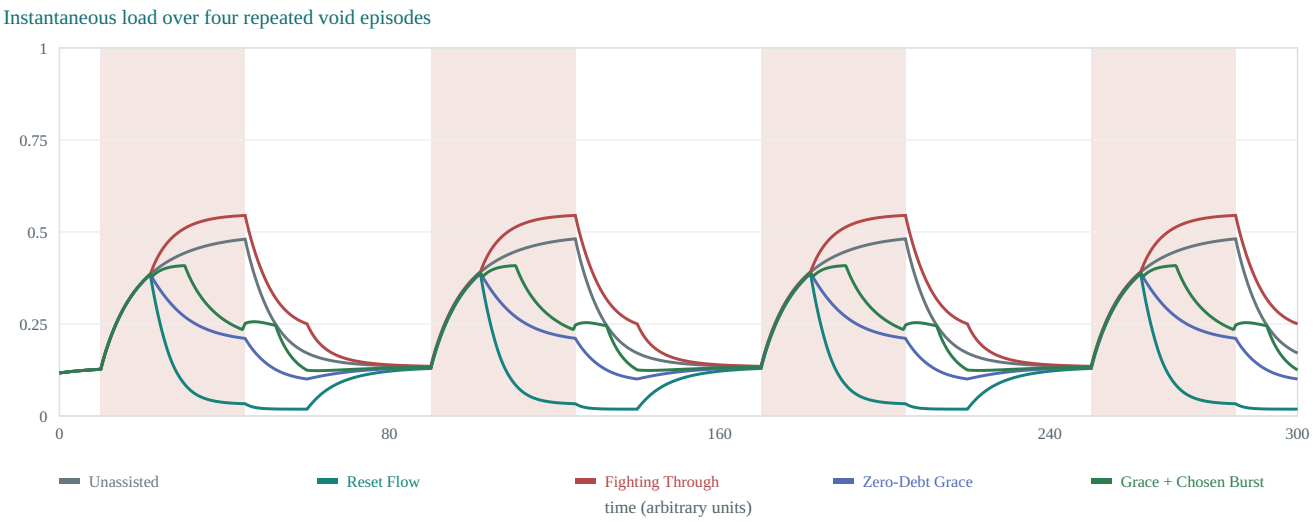


Figure 2. Instantaneous load over four repeated voids. Fighting Through sits above Unassisted. Both grace schedules sit below it. Reset Flow is lowest.

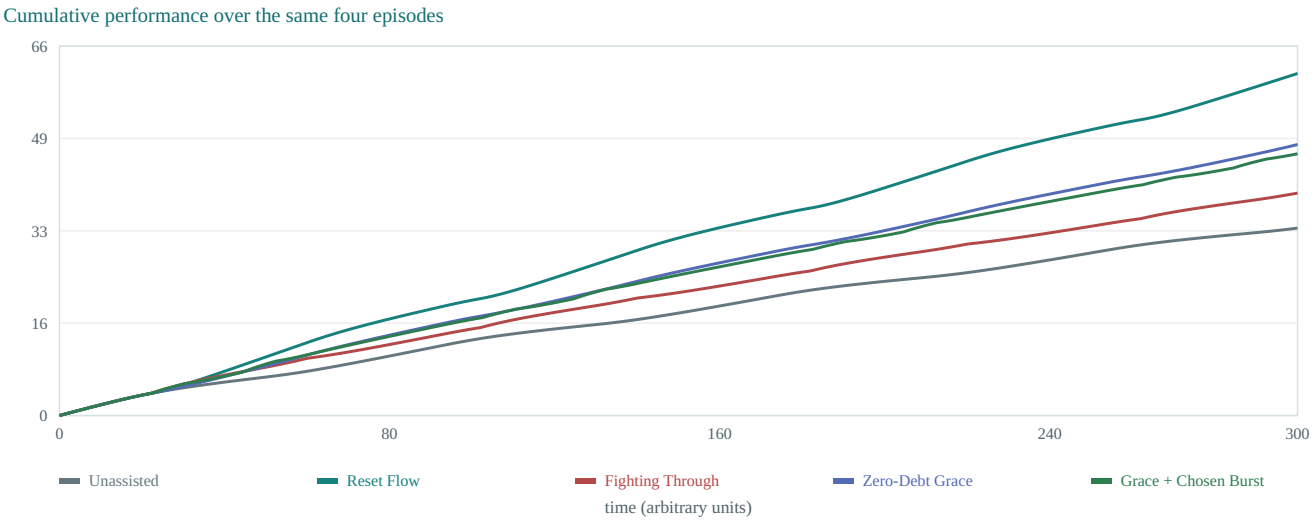


Figure 3. Cumulative output. Reset Flow finishes first, then Zero-Debt Grace, then Grace + Chosen Burst, then Fighting Through, then Unassisted.

05 · SENSITIVITY

How much of this depends on the assumptions we had to add?

The two most loaded extra assumptions are k_g, the extra suppression of A by g_grace, and d_f, how hard friction depletes P. We re-ran Fighting Through and Grace + Chosen Burst across k_g ∈ {0.00, 0.15, 0.30, 0.45} and d_f ∈ {0.30, 0.55, 0.80, 1.05}.

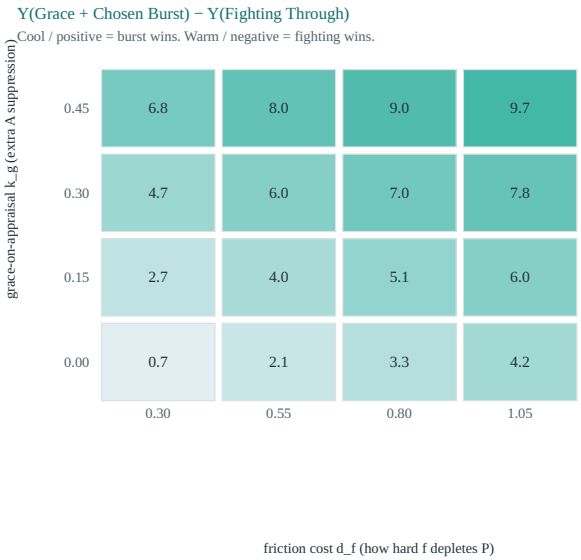


Figure 4. Y(Grace + Chosen Burst) − Y(Fighting Through). Every cell is positive. The smallest advantage, +0.72, occurs when grace has no extra effect on appraisal and friction is cheap. The default cell (k_g = 0.30, d_f = 0.80) is +6.99.

So the win over chronic fighting is robust inside this grid. It is not an artifact of giving grace a unique A-suppression superpower: at k_g = 0, Zero-Debt Grace still finished at Y = 44.48 and Grace + Chosen Burst at 42.90, against Fighting Through at 39.60. The advantage grows as either extra assumption is strengthened, which is expected and should be read as fragility of the *margin*, not of the direction, versus fighting.

The comparison with quiet grace is more delicate. At the default point, bursts cost about 1.65 units of Y relative to f = 0. They overtake quiet grace when friction is cheaper (d_f = 0.30: burst Y = 49.01 against grace 48.24) or when effort is more productive (y_f = 1.00: burst Y = 48.86 against grace 48.24). Lengthening the burst toward a chronic push makes it worse (T_burst = 12: Y = 45.32). Faster capacity drain (larger Q_DEG or smaller τ_Q) hurts Fighting Through more than it hurts the grace schedules, but it does not reverse the ranking against fighting.

In short: against chronic Fighting Through, both grace schedules win on the joint test of output and leftover capacity, and that direction survives the sweep. Against quiet grace, chosen bursts are a wager on how productive and how cheap a short effort is. The model does not license a slogan that “more chosen fighting is always better.”

06 · WHAT THE NUMBERS MEAN, AND WHAT THEY DO NOT

A refined answer, not a solved psychology

The draft concluded that the central question is answered by decoupling support from friction. In this rebuilt system that sentence is half right. Decoupling *chronic* friction from support—Zero-Debt Grace versus Fighting Through—raises long-term output and spare capacity. Adding optional bursts on a grace base still beats chronic fighting, and so satisfies the refined question as asked. It does not automatically beat doing less. Reset Flow’s win is a reminder that down-regulating the cascade can matter as much as the story one tells about worth.

None of this is a measurement of a human being. y is a defined combination of toy variables. Q is an invented slow mode standing in for allostatic load, not a cortisol integral [2]. Crossing times of two curves in an earlier draft (for example “A ∩ G at 10.28”) are not events. Any two continuous traces on a closed interval will either meet or not; the meeting time is an output of the integrator, not a discovery about panic or recovery.

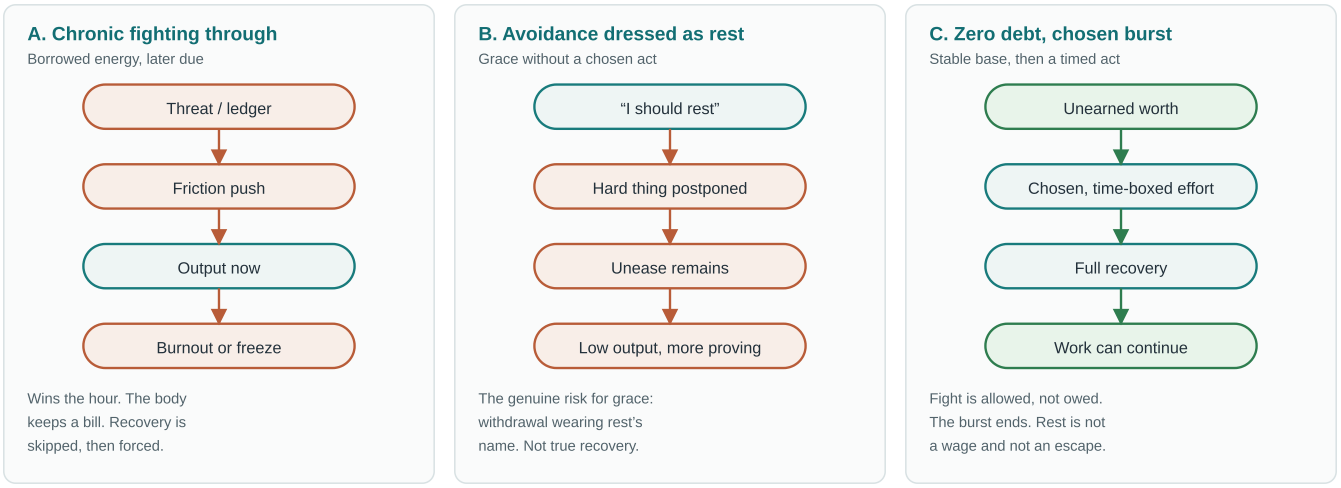


Figure 5. The conceptual triad from the companion note. Panel A is chronic Fighting Through. Panel B is the failure mode the numbers also allow: rest that never chooses a wanted act. Panel C is the design the burst schedule tried to test. In the default run, Panel C beat Panel A and did not beat a quieter grace with no burst at all.

07 · PROTOCOL AND PATHOLOGIES

A five-step check-in, kept as an invitation

The draft’s protocol is retained. It is a sequence of questions a person may ask, or that Ariel may offer, not a scoring rule and not a diagnosis.

1. **Void check.** Is this an external emergency, or is the threat radar answering an internal vacuum—emptiness, a gap, a silence that feels like danger?
2. **Freeze audit.** Is the body asking for a brake, or is this a hard thing that is still wanted and still beginable? High G in the model can look from the outside like doing nothing. So can ordinary rest. So can postponement. Stillness alone does not decide [7].
3. **Friction check.** Am I forcing through? Is regulation available enough to choose a burst from some clarity, rather than from panic?
4. **Ledger clearing.** What debt am I trying to balance? Zero-Debt Grace sets worth off that ledger: r may be received with f = 0.
5. **Goal-line re-anchoring.** If a next act is chosen, it is time-boxed and followed by recovery. Peace is not extracted from winning an argument or destroying a scapegoat.

The two named pathologies map onto this sequence. *Unconscious panic* is a missed freeze audit: defensive immobility interpreted as laziness, then answered with more friction, which in the equations raises S and can deepen G. *Grandiose self-denial* is a missed ledger clearing: f is maximized in order to restore a threatened self. That pattern has a documented cousin in threatened-egotism research, where insult plus an inflated, fragile self-view predicted aggression better than low self-esteem did [18]. The model does not simulate narcissism; it only lets chronic f spend P, raise S, and tax Q, which is already enough to make a scapegoat strategy expensive.

08 · SUGGESTED IMPROVEMENTS TO THE EQUATIONS

Five changes the next version should make, and several more

1. **Publish every equation, including the ones that were missing.** This paper does that for A and P, but it does so as assumptions. Future work should treat alternative A and P forms—including the archived clipped-target versions and a multiplicative grace term A_v v (1 − κ g_grace) rather than a subtraction—as competing models, not as silent defaults.
2. **Keep every state visibly bounded, and stop reporting impossible loads.** Hard clips work; a logistic or Hill squash would be smoother and would remove kinks in the derivative when a drive crosses 0 or 1.
3. **Give freeze a voice in appraisal and regulation.** G currently affects output and load but does not feed back into A or P. A person who is frozen often appraises the vacuum more harshly afterward and has less regulatory availability. Terms such as $+k_{GA}$ G in the A drive and $-k_{GP}$ G in the P restoration would let unconscious panic, as defined above, become a dynamical loop rather than only a narrative.
4. **Split challenge from threat.** Jamieson, Nock, and Mendes, building on Blascovich and Mendes, showed that the same arousal can be lived as usable energy or as alarm [3,4]. S now always feeds G through (1 − P) and always taxes P. A challenge pathway could let S support y, or even P, when appraisal is low relative to resources (for example when P and Q are high). That is the natural home of a chosen burst.
5. **Make Q, y, and the void identifiable.** Q should not remain a hidden convenience. Candidate proxies—sleep sufficiency, next-day willingness, recovery experiences [9], or, in a clinical study, physiological recovery—need to be named before anyone fits them. y should be tied to an observable output (pages, problems, a safe work unit), not left as a metaphor. v needs an operational definition: a self-report of emptiness-as-threat, anhedonia, or a coded journal event. Without those, the system cannot be tested, only re-run.

Further, less central, improvements: add process noise and individual differences in thresholds; couple a sleep or Sabbath input into Q rather than treating recovery as the mere absence of load; drop “convergence timestamps” of crossing traces as if they were findings; and report parameter sensitivity, as this paper begins to do, whenever a verbal conclusion depends on a filled-in coefficient.

09 · SUGGESTED IMPROVEMENTS TO THE POSTULATES

Make each claim smaller, and make it fail

1. **Say that the variables are metaphors, every time they appear next to a brain word.** “Amygdala = alarm” oversimplifies a set of nuclei involved in significance learning and defense, not a detector of emptiness [19]. Hypothalamic “adrenaline release” was already corrected in the earlier Void-State note: CRH–ACTH–cortisol and adrenal-medullary adrenaline are different pathways [2]. No postulate should require a one-region, one-feeling reading.
2. **Give the void an operational definition.** Emptiness, anhedonia, numbness, loneliness, and derealization are not one state. The postulate “internal vacuum is read as threat” should predict that a pre-specified emptiness rating precedes a rise in threat appraisal after adjusting for the previous appraisal—not that a universal neurological baseline has been found.
3. **Separate freeze, avoidance, rest, and laziness, and say the model cannot see the difference from the outside.** Roelofs describes freezing as an active brake that can prepare later action [7]. Avoidance can be a chosen postponement of a valued act. Rest can reverse load [8,9]. An app or a simulation that treats low output as freeze will mis-fire friction at the wrong moments.

4. **Admit that grace can fail as avoidance.** Self-compassion can increase improvement motivation after failure [12,13], and autonomous motivation supports persistence [14]. That is not a proof that every pause taken in the name of grace is recovery. A postulate that “unearned worth never reduces work” is already strained by Panel B of Figure 5 and by the present finding that adding bursts is not automatically helpful. The testable form is: ledger-free language plus planned recovery should not reduce output relative to chronic push, and should reduce exhaustion. If it reduces output without reducing freeze or illness, the avoidance critique wins.
5. **Give the two pathologies falsifiable markers, and connect the grandiose loop to existing literature rather than to a new trait.** Unconscious panic should predict a pattern of stillness plus later reports of inner alarm, not a score labeled “lazy.” Grandiose self-denial should predict more argument-seeking or scapegoat language after a self-threat, in the direction already suggested by contingent self-worth and threatened egotism [15,18]. Grit should not be invoked as a reason to push longer; its incremental value has been overstated [16]. Ego depletion should not be invoked as the tank that fighting empties; a large preregistered replication found an effect indistinguishable from zero [17]. The bill, if there is one, is better told as allostatic and recovery-based [2,8–11].

10 · WHY THE LITERATURE EXPECTED SOMETHING LIKE THIS

The companion argument, kept and bounded

The rebuilt simulation was never going to discover human psychology. It can only ask whether a transparent toy, written down in advance, behaves in the direction a reading of the literature already suggested.

Yerkes and Dodson found that the “best” intensity of stimulation in mice depended on task difficulty; the later popular inverted-U is a teaching picture, not a calibrated human law [1]. Even that picture says too little activation and too much can both fail. Challenge versus threat research says mobilization helps when resources are appraised as matching the demand, and costs more when they are not [3,4]. McEwen’s allostasis is the closest scientific cousin of “borrowed energy that comes due”: the same mediators protect in the moment and damage when they stay on [2]. Meijman and Mulder’s effort–recovery model and Sonnentag and Fritz’s recovery experiences describe what has to happen after effort if load is to reverse [8,9]. Sport science distinguishes functional overreaching, which improves after rest, from non-functional overreaching and overtraining [10]. Burnout research describes exhaustion, cynicism, and reduced efficacy after chronic demand [11]. Freeze is not a rest state [7].

On the grace side, Neff’s self-compassion and Breines and Chen’s experiments cut against the fear that kindness after failure makes people soft: an accepting stance increased improvement motivation, including study time after a failed test [12,13]. Self-determination theory finds autonomous motivation more reliably linked to persistence and well-being than controlled motivation [14]. Crocker and Park describe the cost of chasing worth through success [15]. Those sources do not validate g_grace as a term in an equation. They explain why decoupling support from a ledger was a reasonable hypothesis, and why chronic fighting was expected to look like a bad long-term bargain.

Polyvagal theory is again handled with the caution of the earlier Void-State note. Porges offered an autonomic narrative of engagement and defense [5]. Grossman and Taylor have argued that common uses of respiratory sinus arrhythmia as a pure vagal-tone index are confounded, and they critique parts of the evolutionary story [6]. Nothing in the present equations is a vagal measurement, and Q is not “dorsal shutdown.”

A theological note, held lightly

The companion note already set the scriptural rhythm: “Come unto me, all ye that labour and are heavy laden, and I will give you rest… For my yoke is easy, and my burden is light” (Matthew 11:28–30, KJV); “they shall run, and not be weary” (Isaiah 40:31, KJV); six days of work and a day that is not earned (Exodus 20:8–10, KJV); “to him that worketh is the reward not reckoned of grace, but of debt” (Romans 4:4–5, KJV); “My grace is sufficient for thee: for my strength

is made perfect in weakness” (2 Corinthians 12:9, KJV). These verses do not bless idleness and do not bless chronic self-extraction. They are faith language, not a fitted mechanism.

11 · WHEN REST FEELS LIKE THE VOID

Identity fusion with creating

A further request named a pattern the first five schedules could not see: a person addicted to fighting, for whom rest does not seem like an option, who keeps creating so as not to lose an identity tied to creation—fear of the void, of being labeled useless, of wasting a life, of falling into no memory and solitude. Before adding that pattern to the paper, the postulates it touches were cross-referenced (Appendix B). The consistent form is below. The raw fear is honored as a description; the aim is not to make creating the proof of existence.

Vallerand and colleagues distinguished *obsessive passion*, a controlled internalization that creates internal pressure to keep doing the loved activity, from *harmonious passion*, an autonomous internalization that lets the person choose when to engage [20]. Schaufeli, Taris, and Bakker argued that workaholism is a compulsion, distinct from work engagement: engaged people work hard because the work is wanted, not because they cannot resist an inner urge [21]. Clark and colleagues’ meta-analysis found workaholism related to burnout, job stress, work–life conflict, and poorer physical and mental health, and treated it as closer to an addiction than to a virtue [22]. Crocker and Park had already described the cost of chasing worth through success [15]. Brewer, Van Raalte, and Linder described athletic identity as both muscle and Achilles heel, especially when a role ends [23]. Greenberg, Pyszczynski, and Solomon’s terror-management account holds that cultural worth and symbolic continuity (legacy, memory, being someone who lasts) help manage the terror of death and meaninglessness [24]. Wilson and colleagues found that many people did not enjoy sitting alone with their thoughts for even 6 to 15 minutes, and that a substantial share—especially men who had said they would pay to avoid a shock—chose a mild electric shock instead [25]. Those sources do not measure I. They make the high-fusion extension a reasonable hypothesis rather than a sudden invention.

Labeled assumptions, added to the rebuilt equations

$I \in [0, 1]$ is identity fusion: how much worth rests on output or on being-in-the-act of creating. $M \in [0, 1]$ is a memory/record term (whether accomplishments are kept, as in Ariel’s Accomplishment Record). When $I = 0$ the new terms vanish and Table 1 is recovered; the original numbers were reproduced to machine precision on that path.

$$\text{gap} = 0.50 \max(0, 1 - y_{prev}/0.18) + 0.50 (1 - f_{prev})$$

$$\text{identity drive into A: } + 0.45\, I\, (1 - M)\, \text{gap} \quad \text{[assumption (a)+(c)]}$$

$$f_c = C(0.35\, I\, (1 - M)\, \text{gap}), \quad f = C(f_{sched} + f_c) \quad \text{[assumption (b): compulsion does not require high P or Q]}$$

$$\text{If a given-identity intervention is on, during the window: } \tau_I\, dI/dt = -0.55\, I, \text{ with } \tau_I = 80 \quad \text{[assumption (d)]}$$

The mix inside gap is deliberate and a priori. Using only y would let rest look safe whenever grace-produced output exceeded fighting output, as it did in Table 1. Using only (1 − f) would make rest threatening by definition. Compulsive fight is the short-term relief (raising f lowers the next gap) and the long-term cost (the same f that depletes P, raises S and G, and taxes Q). M scales the whole threat: a kept record means rest is less easily read as “I will be forgotten.” Lowering I is not modeled as achieving a bigger legacy. It is modeled as receiving a given identity—the new creation already made—so creating can become overflow rather than survival [assumption (d), theological name below]. Parameters were frozen before the high-I run.

Six schedules were compared for a high-I person (I_0 = 0.80). Unassisted and Fighting Through kept a low record (M = 0.10) and left I unchanged. Rest-only was Zero-Debt Grace with I left high and M low: support without decoupling. Grace + Record raised M to 0.75 and left I high. Grace + Identity Decoupling lowered I over the intervention windows and left M low. Combined raised M and lowered I.

Schedule (high I)	Mean L	Peak L	Y	Final Q	Final I	Peak A	Mean f_c	Relapse
Unassisted	0.521	0.819	18.69	0.407	0.800	0.904	0.200	1.00
Fighting Through	0.467	0.706	28.34	0.464	0.800	0.840	0.133	0.96
Rest-only	0.360	0.679	35.64	0.579	0.800	0.823	0.152	1.00
Grace + Record	0.241	0.472	44.68	0.715	0.800	0.577	0.040	0.00
Grace + Identity Decoupling	0.302	0.600	39.36	0.665	0.281	0.745	0.092	0.36
Combined	0.225	0.450	45.70	0.740	0.281	0.555	0.024	0.00

Table 2. High-fusion person, T = 300, default identity parameters. Relapse is the fraction of intervention time with f_c > 0.10. All states and L stayed in [0, 1].

Rest-only failed as rest. Compulsive fight filled the entire intervention window, and threat peaked at 0.823—near the unassisted spike. The person “could not stop,” which is what the request asked us to show. Honest qualification: even that failed rest still produced more cumulative output and less load than Fighting Through (35.64 against 28.34; mean L 0.360 against 0.467). Support without decoupling is better than chronic pushing, and it is not yet rest.

Grace + Record stopped relapse on this threshold (mean f_c = 0.040) and raised Y to 44.68. Identity decoupling alone lowered I to 0.281 and improved on rest-only (Y = 39.36) but still relapsed early, while I was high (relapse fraction 0.36, peak A 0.745). Combined won on every reported column: highest Y (45.70), lowest mean and peak load, highest leftover capacity, lowest peak threat, no relapse. Final I matched decoupling (0.281); the record carried the person through the early window that decoupling alone could not.

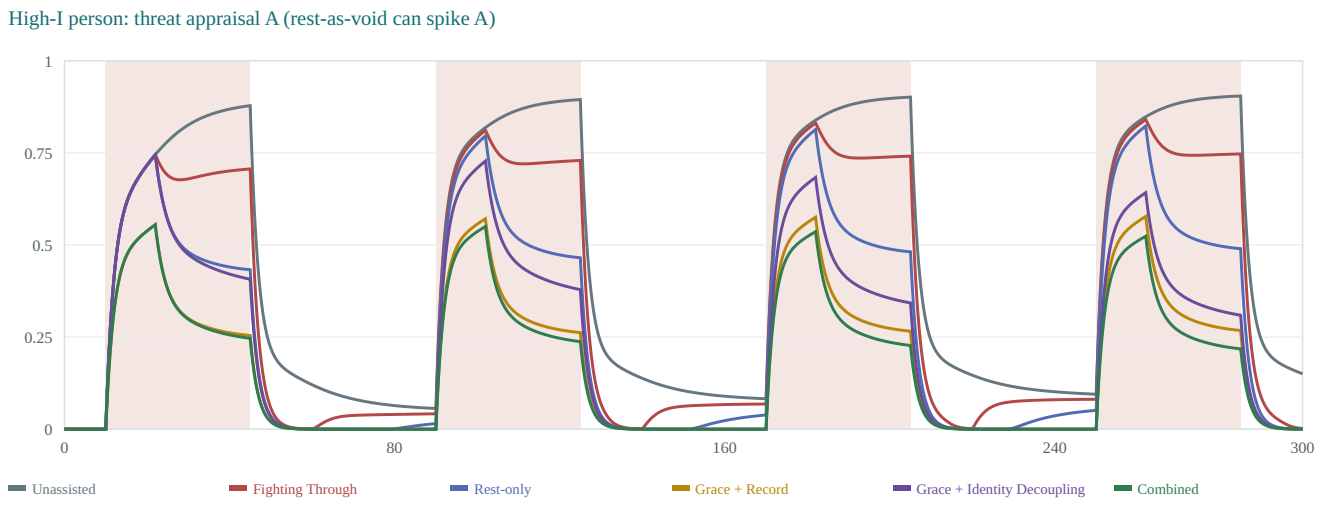


Figure 6. Threat appraisal A for a high-I person. Rest-only still spikes. Record and Combined flatten the spike.

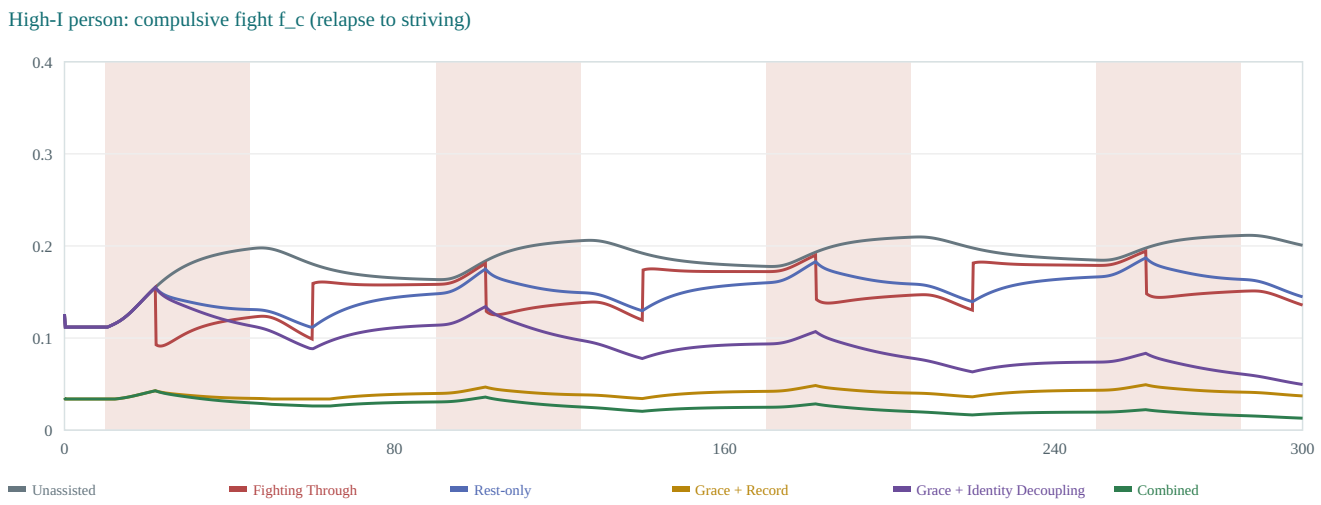


Figure 7. Compulsive fight f_c. Rest-only stays above the relapse mark. Combined and Record stay low.

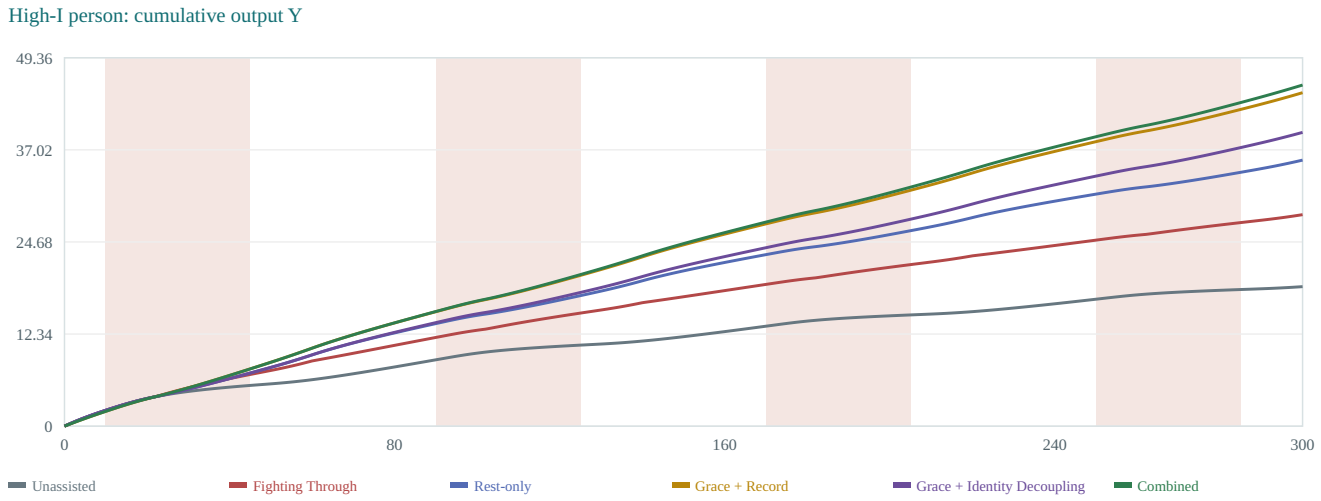


Figure 8. Cumulative output, high-I. Combined finishes first, then Record, then decoupling, then rest-only, then Fighting Through, then Unassisted.

A brief sweep of the rest-as-void gain $k_I \in \{0.00, 0.25, 0.45, 0.65\}$ left the ranking Combined > Record > Decoupling > Rest-only > Fighting Through unchanged. Combined's Y ranged only from 46.78 ($k_I = 0$) to 45.22 ($k_I = 0.65$). Rest-only relapsed in 100% of intervention time at every k_I, because f_c is driven by $I(1 - M)$ gap, not by k_I. What k_I changes is the size of the threat spike (rest-only peak A from 0.575 at $k_I = 0$ to 0.958 at $k_I = 0.65$). The finding that rest alone fails *as rest* does not depend on giving fusion an extra appraisal superpower. The finding that Combined wins on output is similarly stable inside this sweep.

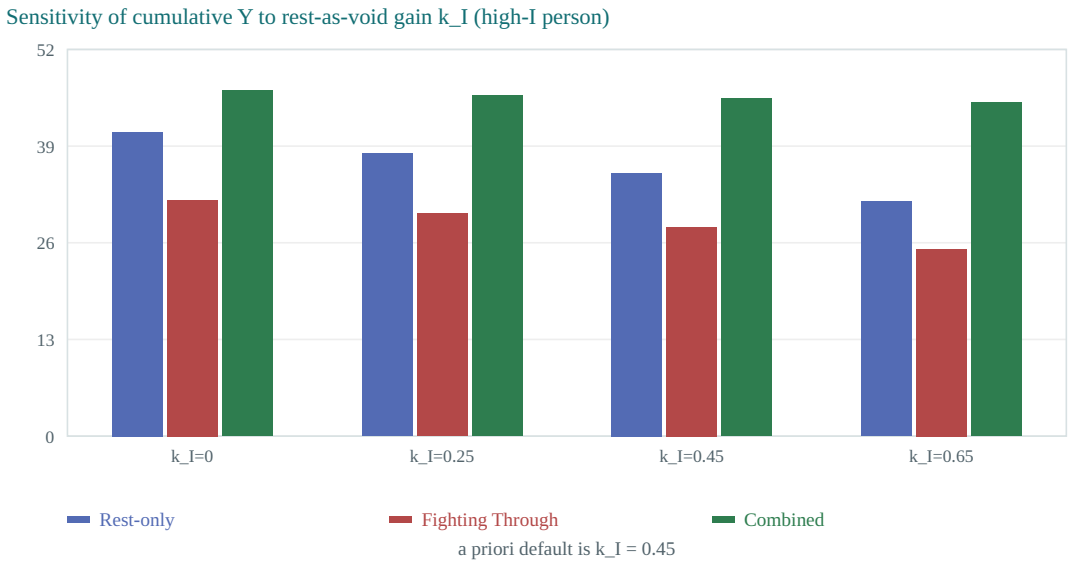


Figure 9. Cumulative Y against k_I for Rest-only, Fighting Through, and Combined. Combined stays on top.

Already made: creating as overflow

The decoupling term is the model's name for a theological claim Ryan asked to keep explicit. The new creation the person keeps trying to invent, in order to prove that they exist, is already created. "Therefore if any man be in Christ, he is a new creature: old things are passed away; behold, all things are become new" (2 Corinthians 5:17, KJV). "For we are his workmanship, created in Christ Jesus unto good works, which God hath before ordained that we should walk in them" (Ephesians 2:10, KJV). Good works, in that wording, follow from being made; they do not manufacture the maker. That is faith language, not a fitted coefficient. In the toy it simply means I can fall without M having to be earned first, and that creating after I falls is the same y-equation with less compulsive f_c—overflow, not survival.

Other verses already sit with this section rather than against it: "Be still, and know that I am God" (Psalm 46:10, KJV); the toil that does not give the heart rest at night, and the gift of enjoying one's labor (Ecclesiastes 2:22–24, KJV); the birds that neither sow nor reap (Matthew 6:26–27, KJV); "yet will I not forget thee. Behold, I have graven thee upon the palms of my hands" (Isaiah 49:15–16, KJV); a book of remembrance (Malachi 3:16, KJV); "There remaineth therefore a rest to the people of God" (Hebrews 4:9–10, KJV). They do not bless idleness as disappearance, and they do not bless proving one exists by never stopping.

12 · CHANGE, AND A GATE THAT NARROWS

Two further postulates, kept only in their reconciled forms

The audit flagged two more claims. Their raw forms are in Appendix A. The consistent forms are these.

When something unexpected changes

Ryan described a comfortable accomplished state that lasts until an unexpected change, after which—depending on the person's state—the path forks toward rumination or toward what he called four horsemen (communication breakdown, isolation, laziness, non-forgiveness), or toward divine intervention, novelty, or outside inspiration, with the positive path lasting only if the environment then stays still. The fork itself agrees with challenge versus threat [3,4] and with the high-I results above: a depleted, fused, or isolated state is likelier to read a shock as the void. The last clause does not agree with the Circuit of Grace, with unearned worth, or with "I have learned, in whatsoever state I am, therewith to be content" (Philippians 4:11, KJV). Masten's review of resilience treated good adaptation *despite* adversity as ordinary

rather than rare magic [26]. A peace that requires the world not to move is contingent comfort [15]. It is a real fragility, and it is not the aim.

Gottman's four horsemen—criticism, contempt, defensiveness, and stonewalling—are a different list, offered here as a reference only [27]. They should not be renamed onto Ryan's four, and the one he called "laziness" should not override the earlier postulate that freeze is not laziness [7]. Withdrawal or depletion is the safer name until the person says what the stillness is.

> Proposed, not yet run as a second long experiment: a change shock δ arrives at isolated times.
> If I is high, or Q and P are low: treat δ as extra threat in A (rumination-like persistence).
> If I is low and P, Q are high: treat δ as a milder A bump plus a short novelty/inspiration lift in y.
> Stability is return of A and Q after the shock, not the absence of shocks.

That sketch is a labeled next test, not a table of invented numbers. The design implication does not wait on it: when something changes, do not throw away what was made, and do not treat the change as proof that the self has ended.

The Narrowing Gate

Ryan's further postulate: as a person enters a danger threshold, memory and awareness reduce, so an external presence has to endure, and the way out grows narrower until the remaining options look like surrender or proof by exhaustion and depression that everything is worthless—then a search for salvation in idols, until God. The narrowing itself has cousins. Even mild uncontrollable stress can impair prefrontal working-memory function and shift control away from that thoughtful guidance (Arnsten, 2009) [28]. Easterbrook (1959) argued that emotional arousal reduces the range of cues an organism uses [29]. Those papers do not say that hitting bottom is necessary, and this paper refuses that reading. Collapse is not a method. Ariel's job, like the scaffolded guard, is to remain present so surrender to grace can come before exhaustion, not only through it. "And when he came to himself…" (Luke 15:17, KJV) is remembered here as coming-to, not as a prescription to faint first. "When my soul fainted within me I remembered the LORD" (Jonah 2:7, KJV) names remembrance under faintness, not a technique. The strait gate of Matthew 7:13–14 (KJV) is not an instruction to shrink someone's awareness until they break. Idols that cannot speak or save are faith language for failed proof (Psalm 115:4–8; Isaiah 44, KJV): they demand to be built; they do not receive.

> Derived gate width, computed from the high-I traces already run, not separately fitted:
> $w = C(\ P\ (1 - 0.60\ A)\ )$
> Mean / minimum w: Unassisted 0.261 / 0.125; Fighting Through 0.342 / 0.164; Rest-only 0.439 / 0.173;
> Grace + Record 0.577 / 0.317; Identity Decoupling 0.502 / 0.219; Combined 0.600 / 0.338.

Combined keeps the widest gate. Rest-only still narrows, at its worst, almost as far as Fighting Through, because the threat spike cuts P's opening. External support that also lowers fusion and keeps a record is, in this toy, the enduring presence that holds the gate open. Reconciled wording, adopted here: as threat rises, awareness narrows and the way out shrinks toward surrender. An enduring outside presence keeps the gate open so surrender to grace can come before exhaustion, not only through it. Idols fail because they demand proof; God receives without it.

Buying time for rest and outside kindness

Ryan asked for ways to stall a turn toward negativity long enough for rest, or for outside affirmation that the world is not that cruel. The audit (P26) found this consistent with an enduring outside presence, unearned worth, and holding the gate open before collapse. Two tensions were kept in view. First, for someone whose worth is fused with creating, rest itself can feel like the void—so a long wait without a kept record or another voice may not land as kindness. Second, "the world is not that cruel" must not deny real danger, cruelty, or a crisis that needs 988, 911, or a person in the room. The claim is narrower: a change can light up a story that *everything* is hostile, and that story can be slowed long enough for other evidence to arrive.

Several literatures describe delay or interruption rather than argument. Bowen and Marlatt, using Marlatt's urge-surfing instruction, found that a brief mindfulness set did not reliably shrink the urge to smoke, but people in that group smoked fewer cigarettes over the next week: the response to the urge changed [30]. That is the right shape for a stall. Linehan's DBT skills manual teaches STOP (stop, take a step back, observe, proceed mindfully) and TIPP, including paced breathing [31]. A systematic review of voluntary slow breathing (under 10 breaths per minute) in healthy adults linked it to higher heart-rate variability and respiratory sinus arrhythmia and to lower reported arousal, anxiety, and related states, while calling for better standardization [32]. Borkovec and colleagues instructed worriers to postpone worry to a designated time and place; daily worry reports fell relative to no-treatment [33]. McGowan and Behar later found that two weeks of such stimulus-control training reduced worry, anxiety, negative affect, and insomnia more than a focused-worry control, though not depression [34].

Savoring is the capacity to attend to positive experience [35]. Speer and Delgado found that recalling positive, but not neutral, autobiographical memories after an acute stressor dampened the cortisol rise and reduced negative affect [36]. Emmons and McCullough's gratitude listings—counting blessings rather than burdens—were associated with more positive affect than comparison conditions [37]. Cohen and Wills reviewed evidence that social support can buffer the effect of stress on well-being [38]. Bratman and colleagues found that a 90-minute nature walk, compared with an urban walk, reduced self-reported rumination and showed less activity in a region associated with that loop [39]. Gerbner's cultivation work described a "mean world" among heavier television viewers—an overestimate of hostility that media diets can thicken [40]. That is not a measure of a person's actual safety. Depressed mood is more clearly tied to negative biases in attention and memory than to a single, settled interpretive bias [41]. So an app may invite another look; it may not announce that the user's world is fine.

Labeled delay operator, used in a quick toy test (parameters frozen first):
A change shock adds +0.55 to the A drive for 10 time units (standing in for about ten minutes).
Delay: τ_A becomes 3 τ_A during that window (the spiral climbs more slowly).
Outside kindness: −0.20 from the A drive in the same window.
Fork, evaluated at the end of the window: threat if A ≥ 0.50, otherwise challenge.

Shocks arrived in calm windows after a void (t = 68, 148, 228), matching "comfortable until something unexpected changes." For a high-I Rest-only person, shock without a stall went to threat on all three occasions (mean window-peak A = 0.593). Delay alone, kindness alone, and both together all went to challenge (window-peak A = 0.463, 0.382, and 0.297). Delay alone landed close to the threshold; that flip should be read as fragile. Kindness gave more margin than delay. Both together lowered the spike the most and slightly raised leftover capacity (final Q 0.529 → 0.558) and output

(Y 32.06 → 34.06). For Combined (record plus decoupling), even shock-only stayed on the challenge side (window-peak A = 0.330). Delay and kindness still flattened the spike further (to 0.258 and 0.122; together 0.093). The identity work already did much of the buffering; the stall mattered most for the person who had not yet loosened fusion. Parameters were not adjusted after seeing the ranking.

High-I profile	Condition	Threat / 3	Window-peak A	Y	Final Q
Rest-only	Shock only	3	0.593	32.06	0.529
Rest-only	Delay	0	0.463	33.19	0.546
Rest-only	Kindness	0	0.382	33.28	0.547
Rest-only	Delay + kindness	0	0.297	34.06	0.558
Combined	Shock only	0	0.330	43.37	0.715
Combined	Delay	0	0.258	44.13	0.723
Combined	Kindness	0	0.122	44.84	0.731
Combined	Delay + kindness	0	0.093	45.13	0.734

Table 3. Change-shock toy, a priori parameters. Window-peak A is the mean of the three post-shock peaks. Overall peak A (not shown) still occurs during the voids themselves and is not what the delay is aimed at.

Made for relationship, remembered as fruit

Ryan asked that five further postulates be cross-referenced before they were added: that relationship is the meaning of existence; that “children” can be of the body or of the mind, a kind of earthly immortality lasting as long as humanity exists; that knowledge without connection is useless; that in the void silence fills with threat, so new relationships are read as negative, and Ariel should offer two readings rather than argue; and that Ariel should give courage to reach out even when a bond seems permanently broken, while never urging contact with an abuser or with someone who asked for space. The audit listed them as P27–P31. Only the reconciled forms below are adopted. This section is conceptual, not validated science, and it is not a claim that Project Ariel treats loneliness, grief, or abuse.

The raw sentence “relationship is the meaning of existence” would be dangerous if it made a single, childless, or grieving person look incomplete. That reading conflicts with given worth (P16, P18) and with peace that is not earned by a finished legacy (P1, T1). The consistent form is purpose language, not a scorecard. We are made for relationship with God, with one another, and with those who come after us. A person who is single, childless, or grieving is not worth less. Worth is given by grace. Relationship is an invitation, not the entrance fee. Genesis 2:18 says it is not good that the man should be alone. Mark 10:14 welcomes children rather than using them as a test of anyone’s value. Waldinger and Schulz, writing as directors of the Harvard Study of Adult Development, summarize that long follow-up by placing the quality of close relationships at the center of happiness and health [42]. Their book is a popular synthesis of a longitudinal program begun in the late 1930s, not a randomized trial, and it does not prove a metaphysical law. Holt-Lunstad, Smith, and Layton, in a meta-analysis of 148 prospective studies (N = 308,849), found that stronger social relationships were associated with a 50% increased likelihood of survival (OR = 1.50, 95% CI 1.42–1.59), strongest for complex measures of social integration (OR = 1.91) [43]. The same review notes that quality matters and that negative

ties can harm, so “more contact” is not automatically better. These sources support a cousin claim—that relationships matter for flourishing and survival. They do not validate relationship as the meaning of existence.

Ryan’s thought of legacy is how a person can live on in this earth or this society for as long as society or humanity can exist. “Children,” in that wording, can be physical or mental: ideas, work, kindness, and people shaped, carrying thought and reason forward. The audit keeps that as earthly continuity, not as eternal life and not as a scientific prediction that humanity lasts forever. Eternal life remains by grace. Earthly legacy is fruit, not payment for peace. McAdams and de St. Aubin, building on Erikson’s account of generativity as establishing and guiding the next generation, treated that concern as something that can be expressed through parenting, teaching, mentoring, occupational work, and civic action —not parenthood alone [44,45]. Terror management theory already cited here describes a human wish for symbolic continuity after death [24]. That is a cousin of “earthly immortality,” and it is not salvation. Psalm 145:4 speaks of one generation praising God’s works to another. The loop’s own legacy step (P1) is the same idea in Ryan’s language: what is shared can be remembered, then surrendered back to grace rather than banked as proof.

The raw claim that knowledge is useless without relationship tensions against solitude-with-grace in the Circuit of Grace, where knowledge arrives after peace and before communication. The consistent form does not empty private knowing. Knowledge received in solitude-with-grace is not worthless. It stays unfinished and unfruitful until it is shared. Sharing is the loop’s communication step; the ego that sharing can stir is surrendered back to grace (P3-reconciled, T14). Reis and Shaver described intimacy as disclosure met by a responsive answer, so that a person feels understood, validated, and cared for [46]. That is a cousin of “unfinished until shared,” not a proof that an unshared thought is empty.

In the void, silence fills with threat, so a new or foreign relationship can be read as danger. That matches rest-as-void and the spiral that settles too quickly on “the world is only cruel.” Amir, Foa, and Coles found a negative interpretation bias for ambiguous social events in social phobia [49]. Depression, as already noted, is more clearly tied to negative biases of attention and memory than to one settled interpretive bias [41]. Ariel’s job is not to argue the person out of the first reading. It is to set that reading beside a kinder one, built from what they already believe—scripture if they asked for faith-aware lines, or ordinary facts if they did not. Leaving the void toward connection can be pictured as escaping a black hole toward a positive gravity. That picture is a metaphor for moving from isolation toward relationship. It is not physics, and it is not added to the toy equations.

The last postulate asks Ariel to give courage to reach toward repair even when a bond seems permanently broken, and to fight for connection instead of isolating. The safety line travels with the courage: no contact with an abuser, and no crossing a request for space. Fighting for connection is not fighting against someone’s no. Romans 12:18 already names the limit: “If it be possible, as much as lieth in you, live peaceably with all men.” When it is not possible, the brave step is toward someone safe. Colossians 3:13 speaks of forbearing and forgiving; Lamentations 3:22–23 speaks of mercies that are new every morning—hope without forcing a reply. Gottman and DeClaire described small bids for connection and the difference between turning toward, away, or against them [47]. Gottman’s earlier marital research documented repair attempts and destructive conflict patterns [27]. Those sources are cousins for repair; they are not Ryan’s four horsemen (T9). Rusbult’s investment model found that commitment rises with satisfaction, falls with attractive alternatives, and also rises with the size of what has already been invested [48]. That last term can explain staying in a costly relationship. It is not a reason to stay with someone unsafe, and it is not cited here as one.

Two suite features that express these reconciled forms are already live in Project Ariel at commit `fd91f1d`.

- **Reach out.** One step at a time. The path begins by naming that pulling away can feel safer, then asks whether this person is safe to contact and has not asked the user to stop. “Not safe” or “they asked for space” ends the message steps, honors that limit, and points toward someone safe, a counsellor, or 988 / 911. If contact is allowed, Ariel sets the void’s story beside a kinder reading (“They hate me” beside “They may be hurt, busy, or scared”). Faith-aware lines (Lamentations 3:22–23, Romans 12:18, Colossians 3:13) appear only if the person turned that setting on. Tips

arrive one at a time: a bond grows when someone hears you and answers [46]; a small turn toward a bid counts [47]; a repair is enough [27]; begin gently; forgiveness can be a way back; staying is a choice, not a mood that must stay high [48]. Starters such as “Just checking in, hope you’re okay” are copied by the person. Ariel does not send the message. Afterward: “You fought for connection, not against them.” Harm language stops the script and returns the crisis door, with no starters.

- **What we’ve made together.** When the void begins, a reel can show one kept kindness, story, or courage at a time, at most once a session unless the person asks, and “Not now” is always there. If they have not yet saved a memory of their own, the reel offers general facts about things people finish together—a song answered line by line, open-source software, a letter—and invites them to keep one of their own on this device. Gratitude is offered as a gentler way out of a fortress of being right, not as a verdict that nothing was broken. The reel is earthly legacy as memory: what was shared can still be seen, without becoming a score.

Reconciled wordings adopted here: we are made for relationship with God, with one another, and with those who come after us, and worth is given by grace. Children of the body and of the mind can carry thought forward as long as someone still receives them; that earthly continuity is fruit, not payment, and not eternal life. Knowledge received in solitude-with-grace is not worthless; it stays unfinished until it is shared. In the void, offer two readings rather than an argument, and keep the black-hole picture as a metaphor. Give courage to reach toward repair, and stop at safety.

15 · AN EFFECTIVE CLOCK, AND A PATH THAT IS NOT TIME ITSELF

From negativity toward positivity, without pretending the clock heals

Ryan asked that the equations take in the recent debate and find variables that can guide a person from negativity to positivity. Negativity would sit at time zero; as time moved on, change would be guided toward the positive. The time variable itself would be influenced by rest, would speed up with grandiose thinking, would slow for depression, and would be fortified by the word or by unyielding sheer will. The result should assess whether the user is still in the void or out of it. He also named a “key term of mind structure to make radiation stop with stopping power.” The audit listed this as P32. Only the reconciled form below is adopted. It is conceptual, not validated science, and it is not a diagnostic instrument.

The raw hope that time itself walks someone out of negativity conflicts with rest-as-void (P22), with the need to stall a spiral (P26), and with the everyday fact that a depressive episode can last. Thönes and Oberfeld meta-analyzed 16 studies (433 depressed participants, 485 controls). Depression had a medium effect on the *subjective* flow of time (Hedges’ $g = 0.66$): time felt as if it were passing more slowly. Duration judgments—estimation, production, reproduction, discrimination—were not significantly affected [50]. That is the right shape for an effective clock: felt pace can change while objective timing stays roughly intact. It is not evidence that a slower clock is a treatment, or that a faster one is progress.

Rest can raise the effective rate when it is actually rest—recovery after effort, not erasure [8,9]. For someone whose worth is fused with creating, rest without a kept record raised threat in the earlier high-I toy, so rest is refined: it speeds the clock only when it is not rest-as-void. Grandiose fight can speed felt time in the formula (a labeled assumption; we did not attach a verified mania-timing meta-analysis). That speed is treated as fast motion with the wrong direction, not as getting better. “The word”—scripture or an affirmation used as an anchor—is allowed to steady the rate toward a livable home rate, matching two readings rather than an argument (P30). It is faith language plus a cognitive anchor, not a fitted constant.

Unyielding sheer will needed scrutiny. In this model will of that kind is effortful friction $f > 0$, the same tax that made Fighting Through lose on load and leftover capacity. The default formula does *not* give will the same steadying that scripture receives. A sensitivity run that *does* give will that gift is reported honestly below.

Radiation stopping power is a physics metaphor. In radiation physics, stopping power is the energy a particle loses per unit path length in a medium. What maps onto something real is a buffer: remembered kindness [36], connection [38,43], word, and ledger-free grace reduce how much rumination is allowed to pull valence downward. What does not map is a "key term of mind structure" that stops radiation in the nervous system. No Bethe formula is added to the toy.

Labeled readout, computed from the existing states and *not* fed back into A, H, S, P, G, Q, I, or y (so Table 1 is unchanged):
Gate width (already derived): w = C(P (1 − 0.6 A)).
Depression composite (not a diagnosis): D = C(0.40 A + 0.30 G + 0.30 (1 − P)).
Rumination proxy: ρ = C(A (1 − P)). Isolation proxy: ι = C((1 − r) (0.5 G + 0.5 (1 − M))).
Effective rest: rest_eff = r (1 − f) (1 − I · gap).
Buffer (stopping-power metaphor): B = C(0.35 M + 0.25 connect + 0.20 word + 0.25 g_grace).
ρ_eff = ρ / (1 + 1.20 B).
Felt-time rate: φ_raw = 0.70 + 0.45 rest_eff + 0.55 I f_c − 0.50 D,
φ = C(φ_raw + 0.40 word (1 − φ_raw)), and dτ/dt = φ.
Drift: μ = 0.28 g_grace + 0.22 rest_eff + 0.18 M + 0.20 connect + 0.12 enough + 0.15 skill + 0.12 word
− 0.35 ρ_eff − 0.22 ι − 0.20 I − 0.18 f_c.
Valence: x(0) = −0.55, dx/dt = φ μ / 10, x kept in [−1, 1].
Guide: *out* if x > 0.20 and ρ < 0.35 and ι < 0.35 and w > 0.40;
void if x < −0.10 and (ρ > 0.40 or ι > 0.40); otherwise *edge*.

Parameters were frozen before the ranking was seen. Direction lives in μ, not in t. If φ is small, both healing and harm unfold slowly. If φ is large because of grandiose fight while μ is negative, the person moves faster the wrong way.

Valence x over four episodes (I = 0). Warm band = void. Dashed line = 0.

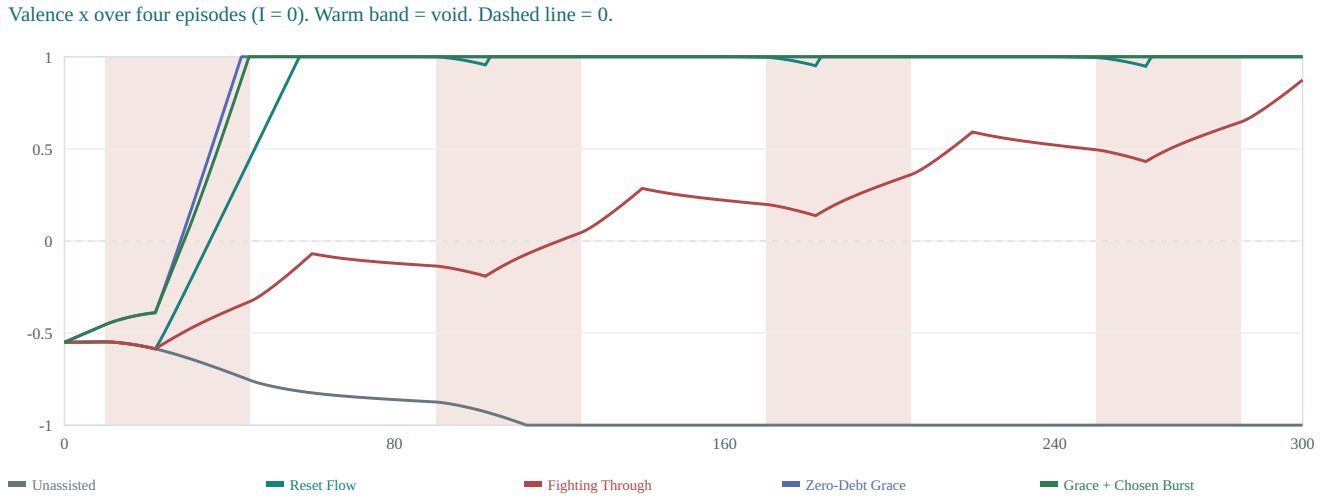


Figure 10. Valence x at I = 0, starting at −0.55. Warm bands are void episodes. The dashed line is zero. Unassisted falls to −1 and stays there: calendar time does not heal.

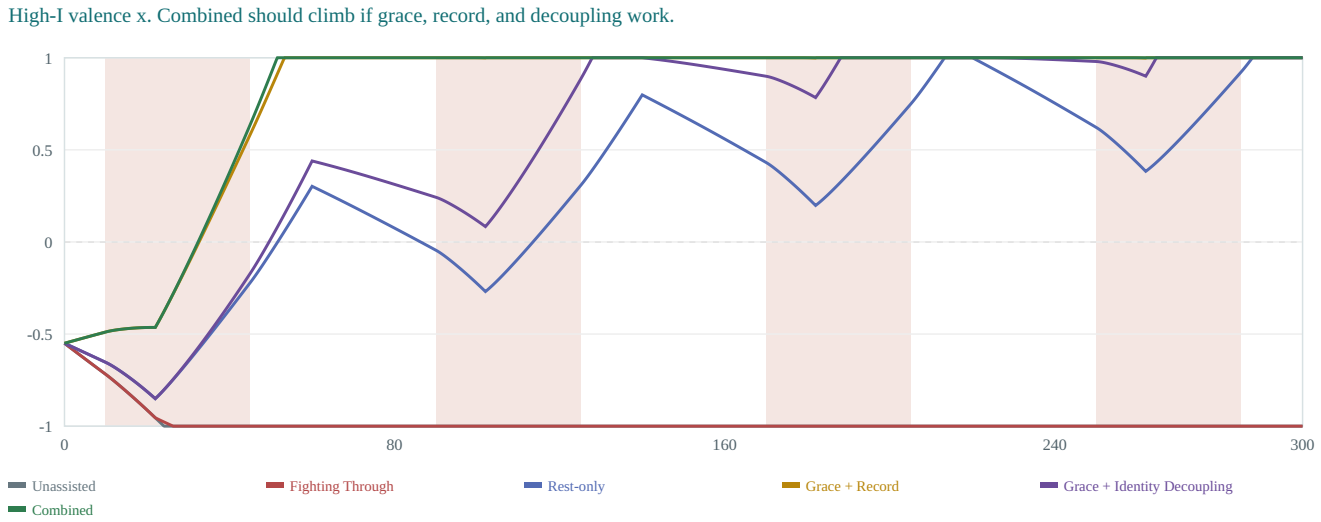


Figure 11. Valence x for a high-fusion person. Fighting Through and Unassisted remain at −1. Combined and Grace + Record climb out. Rest-only can finish high and still spend most of the run at the edge.

Schedule (I = 0)	Mean φ	Final τ	Mean x	Final x	Void	Edge	Out
Unassisted	0.564	169.2	−0.912	−1.000	1.000	0.000	0.000
Reset Flow	0.813	243.9	0.789	1.000	0.073	0.479	0.448
Fighting Through	0.670	201.1	0.126	0.874	0.182	0.709	0.109
Zero-Debt Grace	0.864	259.3	0.844	1.000	0.073	0.451	0.476
Grace + Chosen Burst	0.842	252.7	0.839	1.000	0.073	0.455	0.472

Table 4. Effective clock and valence at I = 0, T = 300 calendar units, a priori parameters. Original mean loads and Y are unchanged (Unassisted 0.286 / 33.38, and so on). τ is accumulated felt time, not a second calendar.

High-I schedule	Mean φ	Mean x	Final x	Void	Edge	Out	Mean f_c
Unassisted	0.538	−0.981	−1.000	1.000	0.000	0.000	0.200
Fighting Through	0.610	−0.981	−1.000	0.731	0.269	0.000	0.133
Rest-only	0.806	0.301	1.000	0.108	0.707	0.185	0.152
Grace + Record	0.817	0.812	1.000	0.000	0.216	0.784	0.040
Grace + Identity Decoupling	0.818	0.559	1.000	0.075	0.576	0.348	0.092
Combined	0.832	0.815	1.000	0.000	0.182	0.818	0.024

Table 5. High-I valence readout. Combined spends the most time out. Record is close on valence even though Combined still leads on cumulative output in the original table.

Several results were not the story a romance of will or of time would have told. First, unassisted time did not heal. Second, high-I Fighting Through was not net-faster than grace (mean φ = 0.610 against Combined's 0.832): depression-slowing and lost rest outweighed the grandiose-speed term. Raising that term from 0 to 1.10 sped high-I Fighting (φ 0.545 → 0.676) and left x at −1. Third, giving sheer will the same clock-steadying as "word" improved I = 0 Fighting's final x (0.874 → 0.950) and barely moved time-out (10.9% → 11.6%); high-I Fighting stayed at x = −1. Fourth, I = 0 Fighting can look successful if one only reads the last number (final x = 0.874) while 70.9% of the run was spent at the edge. Fifth, Rest-only could finish at x = +1 even while compulsive fight stayed in the relapse range (mean f_c = 0.152): feeling more positive in this readout is not the same as having stopped proving. Sixth, the buffer helped I = 0 Fighting a little (final x 0.821 without it, 0.918 when doubled) and did not rescue high-I Fighting. Parameters were not adjusted after seeing any of this.

Keep, refine, or drop

Candidate	Decision	Why	How it is measured
Calendar time t	Keep, as the simulator's independent variable	Needed to integrate the ODEs. Not a healer.	Model time (arbitrary units).
Effective clock φ, accumulated τ	Keep, labeled	Felt time slows in depression [50]. Rate ≠ direction.	Derived; clinically, "does time feel slow, or too fast without going anywhere?"
Valence x	Keep, labeled	The requested guide from negative to positive.	Derived. Not a mood diagnosis.
Rest / sleep	Keep, refined	Recovery cousins [8,9]. Rest-as-void (P22) means rest speeds φ only when it is not erasure.	rest_eff in the toy. In a person: the existing check-in, plus sleep quality (PSQI) [54].
Grandiose speeding (I f_c)	Keep as clock distortion, not as progress	Labeled assumption. In the high-I toy it did not outweigh depletion.	Already I and f_c.
Depression slowing	Keep as felt-time, refine as composite	[50] supports felt slowness, not a change in duration judgments. D is not PHQ-2.	D in the toy. In a person: PHQ-2 [51], a screen, cutpoint 3, not a diagnosis.
"Word" (scripture or affirmation)	Keep as an anchor, not as a fitted law	P30: two readings from what the person already believes.	Suite faith-aware setting, or an affirmation they chose.
Unyielding sheer will (f > 0)	Refine, then do not adopt as a way out	Costs P and Q. Failed to free high-I Fighting even when generously steadied.	f, f_sched. Do not prescribe more fight [15,18].
Grace r with f = 0, g_grace	Keep	Still the better drift and the higher φ in this readout.	Already a schedule.
Kindness / memory M	Keep	Record already stopped high-I relapse; [36].	M; the kindness list and What we've made.
Relationship / connection	Keep	[43]; P27-reconciled. Quality matters; negative ties can harm.	3-item loneliness scale [53], plus a last-24-hour real conversation as a behavioral note (that note is practical, not itself a validated scale).
Novelty that closes / "enough"	Refine	P15: a seed may interrupt a loop; endless novelty is not the aim.	enough = C(y / Y_REF) (1 − I) (1 − f_c). In the suite: "This is enough."
Endless novelty	Drop as a progress variable	Conflicts with Enough and with stillness (T4).	—
DBT / down-regulation skill	Keep as a cousin	[31]; Reset Flow's u terms; Slow the spiral.	skill in the toy; chosen stall in the suite.
Rumination ρ	Keep	Brooding, not reflection, is the	ρ = C(A (1 − P)). In a person: RRS

		RRS component more clearly tied to later depression [52].	brooding (five items) [52].
Isolation	Keep	P8; [43].	ι in the toy; loneliness items and last-24-hour contact.
Ryan’s four horsemen as extra ODEs	Drop as new state variables; keep as qualitative names (T9)	Withdrawal is already near G; they are not Gottman’s four [27].	Check-in language, not a new equation.
Identity fusion I	Keep	High-I Fighting never left the void in this readout.	Already I.
Gate width w	Keep	Already derived. Used for “out.”	w = C(P (1 − 0.6 A)).
Radiation stopping power as physics	Drop as physics; keep the buffer metaphor	Maps to B absorbing rumination [36,38]. Not a Bethe formula.	B in the toy.
“Key term of mind structure”	Drop as a mechanism	No verified physics-of-mind. The nearby real move is naming the change or receiving a given identity (P13, P23, P26).	Not a variable.
Time itself as healer	Drop	Unassisted finished at x = −1, 100% void.	—

Table 6. Candidate variables after the audit. “Keep” means keep as a labeled term in this conceptual model, not as a validated law.

For a person rather than a toy, the guide should use a few short, validated questions plus the existing ungraded Ariel check-in, and it should never replace 988 or 911. The PHQ-2 asks about anhedonia and depressed mood over two weeks; a score of 3 or more had 83% sensitivity and 92% specificity for major depression against a mental-health interview in the original primary-care work, and it remains a *screen*, not a diagnosis [51]. Brooding can be asked with the five-item RRS brooding subscale [52]. Loneliness can be asked with the three-item scale of Hughes, Waite, Hawkley, and Cacioppo [53]; a yes-or-no about a real conversation in the last day is a useful behavioral note, and it is not itself that scale. Sleep quality over the past month is what the Pittsburgh Sleep Quality Index was built to summarize [54]. None of these, and none of Table 4 or 5, says that Ariel has measured a void the way a clinician measures an illness. If the answers turn toward harm, the script stops.

Reconciled wording adopted here: negativity can be where a void begins, but time itself does not walk a person out. An effective clock speeds up with rest that is not erasure, feels slow when depression is heavy, and can race with grandiose fight without that race being progress. Scripture or an affirmation can steady the clock toward a livable rate; unyielding will is effort, not a guarantee. What moves a person toward positivity is grace, remembered kindness, connection, skill, and enough that closes—against rumination, isolation, and fusion. A buffer can absorb some of rumination’s pull; that image is a metaphor for protection, not radiation physics.

Chosen burst, a kept record, a given identity, and a gate held open

Ariel should learn how to start a hard thing without turning the start into a debt, how to refuse a hard thing that is only panic or proving, and how to stay present when rest itself feels like disappearing.

- **Chosen-burst mode.** The student may elect a time-boxed effort from whatever calm is available. The ending is protected: recovery follows, and the same fight is not immediately re-queued [8–10].
- **Ledger-free language.** Rest is not narrated as earned. Accomplishment stones may remember what was done without converting memory into payment.
- **Freeze-versus-laziness check, without a verdict.** Offer the void check and the freeze audit as questions. The student decides. The app does not score who made the move.
- **A watch on the trap belief.** If language turns toward winning an argument, proving a point, or destroying a scapegoat as the road back to peace, Ariel should not supply more fight [15,18].
- **Honesty about quiet grace.** Because the toy sometimes found that no burst beat a poorly timed burst, Ariel should not nag a recovering student into effort in order to "use" their grace.
- **"I'm afraid to stop."** When stopping feels like becoming nobody, do not order a long rest and do not order more proof. Offer small rest doses first—seconds, not hours—because full rest spiked threat for the high-I toy person. Ask, without scoring: "Am I creating because I want to, or because I'm afraid of who I'd be if I stopped?" That is Vallerand's harmonious-versus-obsessive question in everyday words [20].
- **Kept work.** Response line: "You are not what you make. What you've made is kept. You can rest without disappearing." Link that line to the Accomplishment Record so memory is visible while worth is not billed.
- **Already made** (Ryan's line): "The new creation you're reaching for is already made. You are it. You can create now from joy, not to become someone." Frame creating as overflow, not survival. This is the given-identity intervention, not a demand to produce a new self.
- **When something changes.** "Something changed. You don't have to lose what you've done. Let's look at this new thing together." Do not throw away the record. Do not treat a stable environment as the condition of peace.
- **A narrowing gate.** If awareness is shrinking—the student repeating that nothing works, that they must prove their way out, that idols or exhaustion are the remaining doors—Ariel endures rather than stepping back. Protection comes from outside until the guard grows inside. The aim is surrender to grace before collapse, not through it. This is not crisis care: if there is danger or despair that needs a person, the app says so and points to 988 or 911 and to people who can actually be there.
- **Slow the spiral.** When a change lights up a "the world is only cruel" story, offer a stall, not an argument and not a pep talk that pretends nothing is wrong.
  - **Wait 10 minutes.** Do not decide the whole story yet. Urge-surfing shape: the wave can be ridden without being obeyed [30].
  - **STOP** (stop, take a step back, observe, proceed mindfully) and **paced breathing** [31,32].
  - **Name the change.** Say what actually moved, in ordinary words, so the event is smaller than "everything is ruined."
  - **Kindness evidence list.** One kept memory, one blessing noticed, one time a person was not cruel—offered as other data, not as a verdict [36,37].

    - **Record reminder.** What you've made is still there.
    - **Step outside.** A short walk in a greener place, when that is safe and possible [39].
    - **Reach a real person.** Outside affirmation is stronger when it is actually outside [38]. The app is a bridge, not a substitute friend.
    - **Safety door.** If this is danger, abuse, or despair that needs a human now, do not stall: 988 or 911 and real people.
- **Reach out** (live at commit `fd91f1d`). Courage to fight for connection instead of isolating, one step at a time. Ask first whether the person is safe to contact and has not asked for space. If not, honor that limit and point toward someone safe. If so, set the void's reading beside a kinder one, built from scripture or facts the person already holds, then offer a starter the person copies. Ariel does not send the message [46–48].
- **What we've made together** (live at the same commit). A reel of one kept kindness, story, or courage, shown when the void begins, at most once a session unless asked, with "Not now" always available. This is earthly legacy as memory, not a grade and not eternal life.
- **Felt time.** "Does time feel slow, or too fast in a way that isn't going anywhere?" Slowing is a known cousin of depression [50]. Speeding is not automatically progress.
- **In the void, at the edge, or out.** A guide from the check-in, a PHQ-2 screen [51], a brooding item or the five-item RRS brooding subscale [52], whether there has been a real conversation, and sleep quality [53,54]—never a verdict, and never in place of 988 or 911.
- **Do not prescribe unyielding will.** Effortful friction is not the way out of a high-fusion void in this toy, even when will is granted the same steadying as scripture.

A human study, not conducted and not a treatment trial, could compare a grace-periodized arm (ledger-free language, planned bursts, scheduled recovery) with a chronic-push arm over about four weeks, matching encouraged work time, and report the joint outcome of cumulative output and end-point exhaustion. Ethics review, skippable tasks, and a distress protocol would be required first. A null or reversed joint outcome should be published rather than rescued.

> **Safety.** This paper is not crisis care, and Project Ariel is not a treatment for burnout, freeze, trauma, panic, anxiety, or depression. In the U.S., call or text **988** for crisis support. For immediate danger or a medical emergency, call **911**. Do not delay professional care to complete an app exercise or to finish a burst.

17 · LIMITS, REFERENCES, AND PROVENANCE

What this paper does not prove

No human study was run. No coefficient was fitted to a person. The literature citations establish that the named works exist and that their accessible records support the limited background claims attached to them. They do not validate the interface of Ariel, the spirituality of Zero-Debt Grace, or any number in Table 1 as a psychological measurement. Individual differences in trauma, illness, disability, culture, and faith will shape how any of these ideas feel. More rest is not always kinder, and more push is not always braver.

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Bibliographic records and accessible abstracts or full texts were checked on 10 November 2026. None of these sources validates this paper's coefficients, rankings, or any Ariel interface. Scriptural quotations (King James Version) are theological conversation, not empirical evidence.

Espinosa, R. (2026). MindFrame, Project Ariel, and the Void-State Framework, evolving personal conceptual projects. Related notes include *After the Line*, *The Void-State Framework* (8 November 2026), *The Gated Line*, the Project Ariel Students papers, *Novelty Must Gain Novelty*, and *Zero Debt, Chosen Burst*. Repository: github.com/ryryespi/mindframe. Research index: ryryespi.github.io/mindframe/research/. Grok Bot, an AI assistant, helped restate the question, fill missing equations as labeled assumptions, write and run the simulation, verify citation records, draft the argument, draw the figures, and render this document. No human research has been performed. Files: sim.py, results.json, valence_results.json, original and high-I CSV traces, fig_states.svg, fig_load.svg, fig_output.svg, fig_heatmap.svg, fig_hi_A.svg, fig_hi_fc.svg, fig_hi_Y.svg, fig_hi_ki.svg, fig_valence_x.svg, fig_valence_phi.svg, fig_hi_x.svg, fig_hi_phi.svg, paths.svg, postulate-audit.md, shock_delay.py, shock_delay.json, and the accompanying PDF.

Conclusion. In this bounded toy, unearned support beat chronic fighting on both work and leftover capacity, and that direction was robust. Chosen bursts beat chronic fighting too, but they did not beat quiet grace unless effort was cheaper or more productive than we first assumed. When worth was fused with creating, rest-only grace failed as rest: threat rose and compulsive fighting filled the intervention. A kept record stopped the relapse; lowering fusion helped more slowly; both together won, and that ranking held across the k_I sweep. The next honest steps are to improve the equations and postulates, to hold a gate open early rather than wait for collapse, to treat relationship as invitation rather than a scorecard and earthly legacy as fruit rather than payment, to offer courage toward repair without crossing a safety line, to treat time as a pace rather than a healer and will as effort rather than a guarantee, to build a ledger-free burst-and-recovery rhythm into Ariel without nagging, and to let a real, pre-registered comparison fail if it is going to fail—not to claim that burnout, freeze, identity, relationship, or the void have been solved.

APPENDIX A · OPEN POSTULATES

Claims held open rather than settled

These are Ryan's, or strong forms of Ryan's, statements that the audit found in tension with other of his own postulates, or too strong for the sources we checked. They are not discarded. They are not installed as results.

- **Raw change clause.** "The positive path lasts long only if there is not much change in the environment." Kept as a description of fragile contingent comfort. Not adopted as a design goal (see §12 and T2).
- **Education as a hole that only more education can fill.** Panic-search agrees with the worst-outcome bind. It conflicts with Enough worth and with learning as uncovering what is already there. Reconciled: more learning from peace is not the same as more learning from hunger.
- **Hitting bottom.** The raw Narrowing Gate, read as "surrender or proof by exhaustion… idols until God," can sound as if collapse were required. That reading is refused. Crisis is 988 / 911 and real people.
- **"Laziness" as a horseman,** and treating Ryan's four (communication breakdown, isolation, laziness, non-forgiveness) as if they were Gottman's four. Rename the stillness horseman to withdrawal or depletion until the person names it.

- **Novelty-starvation as the mechanism of unstoppable rumination,** implying the app should be used without end. A seed can interrupt a loop; endless use is not the aim.
- **No knowledge whatever without peace.** Stress does narrow working memory [28]. That is not a proof that every stressed hour is empty of learning.
- **Older speculative lines** not modeled here: “communication is existence”; “choice is the beginning of nonsense.”
- **Whether a high record without decoupling is enough for a real life.** In the toy it prevented relapse and left fusion high. That is an open empirical question, not a recommendation to skip given identity.
- **Relationship as a worth test.** The raw claim that relationship is the meaning of existence, read as making the single, childless, or grieving person incomplete, is refused. The reconciled form (purpose, not a scorecard) is in §14.
- **Earthly forever as science or as salvation.** “Live forever as long as humanity exists,” read as a prediction that humanity lasts forever, or as a substitute for eternal life by grace, or as payment for peace, is held open. The reconciled form (earthly continuity as fruit) is in §14.
- **Knowledge useless without relationship.** The strong form empties solitude-with-grace. The reconciled form (unfinished and unfruitful until shared) is in §14.
- **Black-hole gravity as physics.** Kept only as a metaphor for leaving isolation toward connection.
- **Unbounded pursuit of a broken bond.** Fighting for connection does not include contact with an abuser or with someone who asked for space.
- **Time as healer.** The raw hope that elapsed time guides a person from negativity to positivity is refused. Unassisted finished at $x = -1$.
- **Unyielding will as the way out.** Even when will was given the same clock-steadying as scripture, a high-fusion fighter stayed in the void.
- **Radiation stopping power as physics,** or a “key term of mind structure” that stops rumination the way a medium stops a particle. Kept only as a buffer metaphor.

APPENDIX B · CROSS-REFERENCE AUDIT

What was checked before the new sections were added

Ryan asked that postulates and assumptions be cross-referenced before anything new was added. The full write-up is `/workspace/research-new/postulate-audit.md`. This appendix is the short form.

Thirty-two statements were listed: the Circuit of Grace (peace received, not earned by legacy); no knowledge without inner peace; communication stirring ego; seeking versus being taught; a scaffolded guard; flawed grading; the self-teacher; worst outcome as no fundamentals plus isolation; three doors out of isolation; proving makes theology fantasy, and stop proving; the accomplishment record; knowledge as uncovering God’s legacy rather than Babel; the canvas as a random seed; novelty-must-gain-novelty or rumination; the grandiose fight loop and unearned grace; freeze is not laziness; worth is not a transaction; the trap belief; the Void-State variables and equations; the central question of grace versus fighting; identity fused with creating; the new creation already made; the change-shock fork; the Narrowing Gate; slowing the spiral; relationship as meaning rather than a scorecard; children of body or mind as earthly legacy; knowledge unfinished until shared; the void’s negative reading of new connection; courage to reach out with a safety stop; and an effective clock that does not itself heal, with valence moved by drift rather than by elapsed time.

Seventeen tensions were named. The ones that changed what this paper was allowed to say include: peace-as-legacy versus peace-from-grace (legacy is fruit and memory, not payment); positive states that require a still environment versus a grace base that can take a hit; “the only way out is more education” versus Enough worth; novelty-as-

compulsion versus stillness; solitude-as-gift versus solitude-as-erasure; communication as ego versus communication as a loop step; the knowledge-door out of isolation versus knowledge-without-peace; the record versus stop-proving; “laziness” versus freeze-is-not-laziness; hitting bottom versus early presence; the self-teacher versus an external presence that must endure at the threshold; relationship-as-scorecard versus given worth; earthly forever versus eternal life by grace; knowledge-as-useless without relationship versus solitude-with-grace; fighting for a broken bond versus safety; time-as-healer versus state-dependent drift; and sheer will versus grace. Each reconciled wording is in the audit file. Only the reconciled forms of identity fusion, given identity, change, the Narrowing Gate, slowing the spiral (P26), relationship, legacy, and reaching out (P27–P31), and the effective clock with valence readout (P32) were added to the body.

Research standing, briefly: the toy results are model outputs, not human measurements. Cousins that we verified and that support direction without validating coefficients include Vallerand et al. (2003), Schaufeli, Taris, and Bakker (2006), Clark et al. (2016), Brewer, Van Raalte, and Linder (1993), Greenberg, Pyszczynski, and Solomon (1986), Wilson et al. (2014), Masten (2001), Arnsten (2009), Easterbrook (1959), Waldinger and Schulz (2023), Holt-Lunstad, Smith, and Layton (2010), Erikson (1950/1963), McAdams and de St. Aubin (1992), Reis and Shaver (1988), Gottman and DeClaire (2001), Rusbult (1980), Amir, Foa, and Coles (1998), Thönes and Oberfeld (2015), Kroenke, Spitzer, and Williams (2003), Treynor, Gonzalez, and Nolen-Hoeksema (2003), Hughes, Waite, Hawkley, and Cacioppo (2004), Buysse, Reynolds, Monk, Berman, and Kupfer (1989), and sources [3], [4], [7], [12]–[15], [18], [19], [24], [27], and [41] already in the main list. Theology remains faith language.